Resume Project

NYC Yellow Taxi Operations & Revenue Analytics | NYC TLC Public Dataset

Technologies: SQL, PostgreSQL, Python, Power BI, DAX, Excel

- Built an end-to-end analytics solution using the NYC TLC Yellow Taxi public dataset (FY 2025).
- Designed a Star Schema and performed data cleaning and transformation using SQL and Python.
- Developed interactive Power BI dashboards to analyze trip demand, revenue trends, payment behavior, and location performance.
- Created KPIs including Total Trips, Revenue, Average Fare, Trip Distance, Trip Duration, and Payment Distribution.
- Generated business insights to support fleet planning, demand forecasting, and operational decision-making.

🎤 Interview Introduction (30–40 Seconds)

"I completed an end-to-end Operations and Revenue Analytics project using the NYC TLC Yellow Taxi public dataset. I designed a Star Schema, transformed the data using SQL and Python, and developed interactive Power BI dashboards to analyze trip demand, revenue trends, payment behavior, and operational performance. The project focused on providing business insights for fleet optimization and executive decision-making."

Top Interview Cross Questions & Best Answers

Q1. Why did you choose the NYC Taxi dataset?

Answer

> I wanted to work on a large-scale public dataset that represents a real transportation business. The NYC TLC dataset contains millions of real taxi trips and provides sufficient complexity to demonstrate SQL, Python, data modeling, and Power BI skills in an enterprise-like scenario.

Q2. What business problem were you solving?

Answer

> The business lacked a centralized analytics solution to monitor revenue trends, trip demand, payment behavior, operational efficiency, and location-wise performance. The objective was to build a reporting solution that could support strategic and operational decision-making.

Q3. Why did you use a Star Schema?

Answer

> The original dataset is transactional and optimized for data collection rather than reporting. I designed a Star Schema to simplify relationships, improve query performance, and support scalable Power BI reporting with reusable dimensions such as Date, Location, Vendor, Payment Type, and Rate Code.

Q4. What KPIs did you create?

Answer

- Total Trips
- Total Revenue
- Average Fare Amount
- Average Trip Distance
- Average Trip Duration
- Average Tip Amount
- Revenue per Trip
- Revenue per Mile
- Peak Pickup Hour
- Peak Demand Day
- Payment Method Distribution
- Airport Trip Percentage

Q5. What insights did you expect from this project?

Answer

> I wanted to identify high-demand pickup zones, peak travel hours, payment preferences, airport trip contribution, and operational patterns. These insights can help improve fleet allocation, staffing, and customer service.

Q6. How did you calculate Trip Duration?

Answer

> I calculated Trip Duration by subtracting the pickup timestamp from the drop-off timestamp. This metric was used to analyze operational efficiency and identify unusually long or short trips that may require further investigation.

Q7. How did you identify Peak Demand Hours?

Answer

> I extracted the hour from the pickup timestamp, grouped trips by hour, and calculated total trip volume. This helped identify the busiest periods and supported recommendations for fleet planning.

Q8. Why did you include Location Analysis?

Answer

> Taxi demand is highly location-dependent. By combining trip records with the Taxi Zone Lookup table, I analyzed pickup and drop-off patterns by borough and zone, helping identify demand hotspots and operational opportunities.

Q9. What challenges did you face?

Answer

> The biggest challenge was handling a large volume of transactional data and preparing it for analytics. I also needed to ensure consistent relationships between the trip records and lookup tables while maintaining data quality and report performance.

Q10. How would this dashboard help the CEO?

Answer

> The Executive Dashboard provides a consolidated view of trip demand, revenue, operational efficiency, and customer payment behavior. It enables leadership to monitor performance, identify trends, and make data-driven decisions regarding fleet planning, pricing strategy, and operational improvements.

Q11. Why didn't you perform Profit Analysis?

Answer (Very Important)

> The NYC TLC dataset provides fare-related fields such as fare amount, tips, tolls, and surcharges, but it does not include operating costs like fuel, maintenance, driver compensation, insurance, or vehicle depreciation. Without complete cost information, calculating profit would be misleading. Therefore, I focused on revenue and operational performance, which are fully supported by the available data.

Q12. If the company provided cost data, what would you add?

Answer

> If operational cost data became available, I would extend the model by calculating gross profit, profit margin, revenue per vehicle, cost per mile, driver profitability, and route profitability. I would also create financial dashboards to support pricing and profitability analysis.

Interview Closing Statement

"Why should we consider this project valuable?"

> **"This project demonstrates my ability to work with large-scale public data, translate business requirements into an analytical model, design a Star Schema, write SQL for data transformation, perform Python-based exploratory analysis, and build executive Power BI dashboards. More importantly, I approached it as a business problem rather than just a reporting exercise, focusing on delivering actionable insights that could support operational and strategic decision-making."**